

ROLL.NO:240701506

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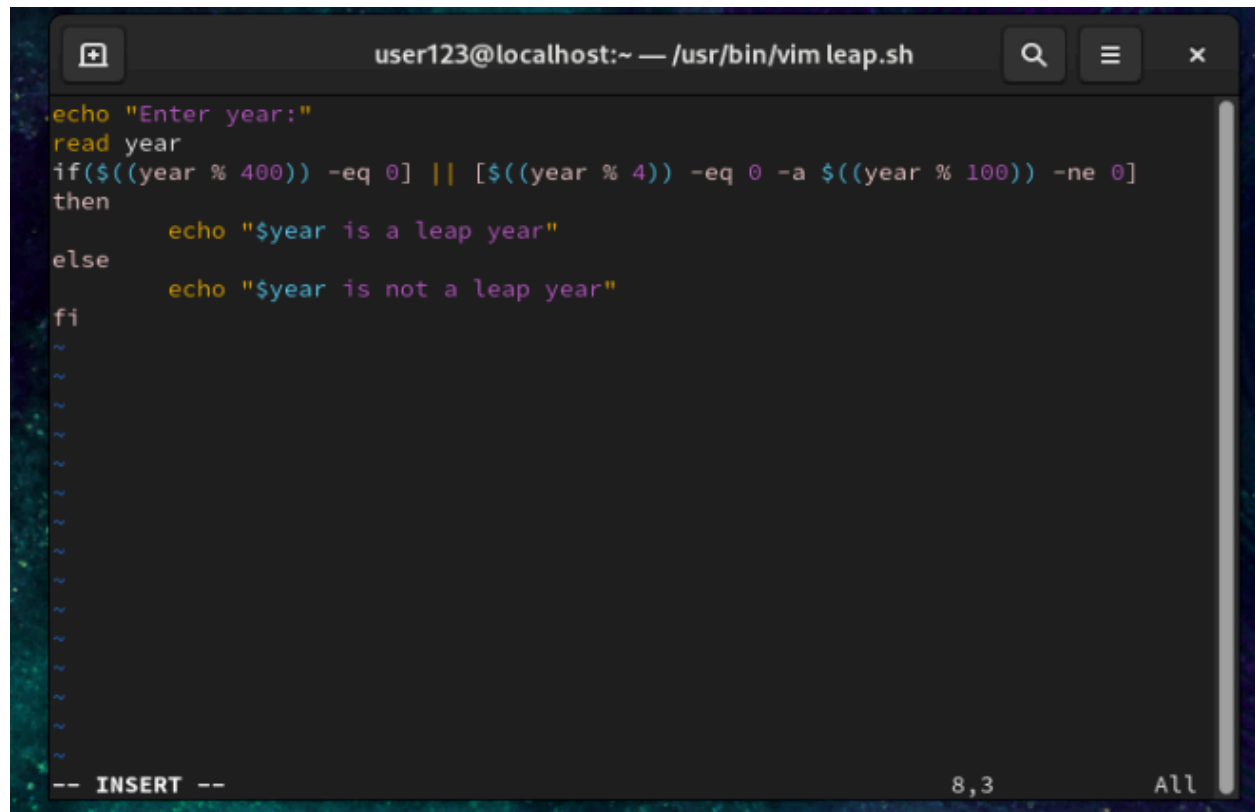
EXPERIMENT - 3 SHELL SCRIPT

1. Shell Script for Leap Year

AIM:

To write and execute a shell script to check whether the given year is a leap year or not.

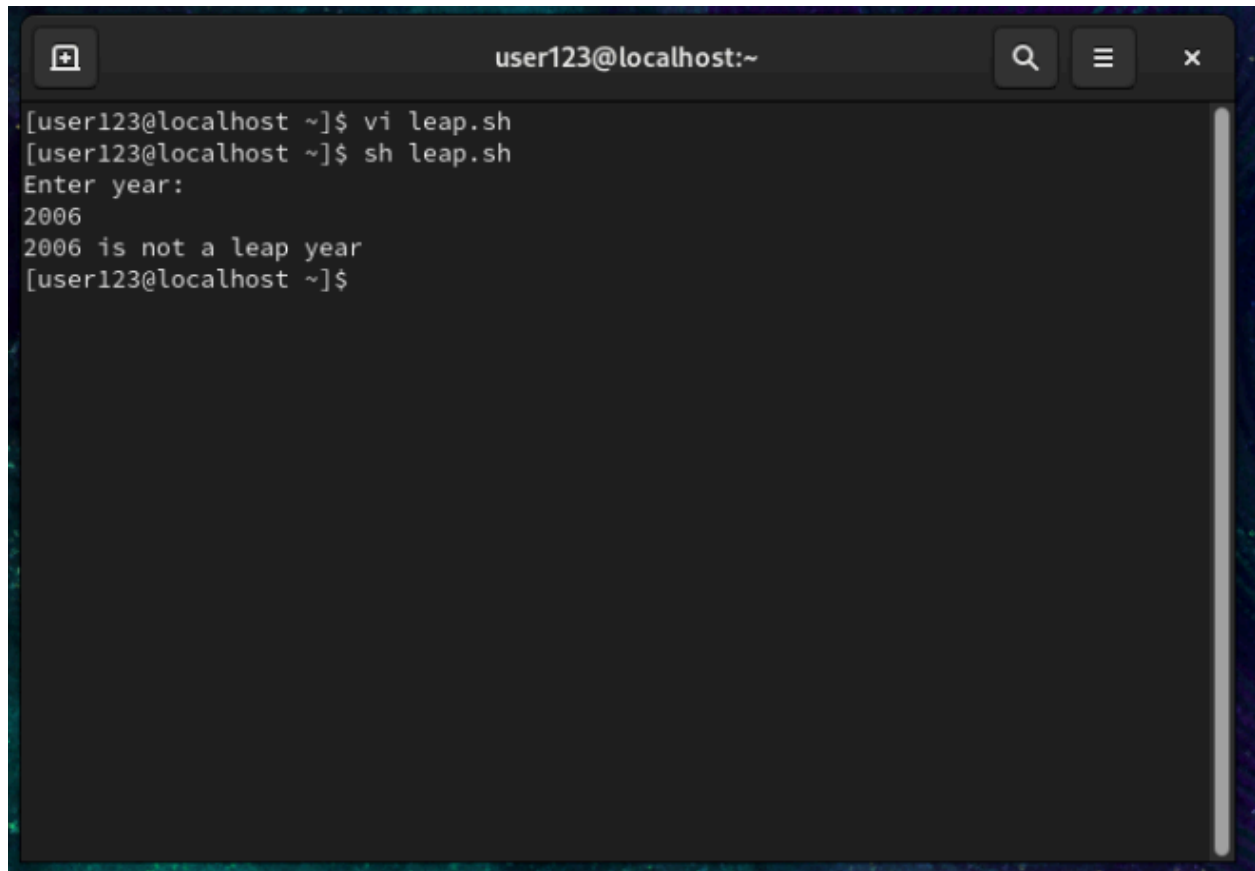
CODE:

A screenshot of a terminal window with a dark background and light-colored text. The window title bar shows 'user123@localhost:~ — /usr/bin/vim leap.sh'. The script content is as follows:

```
echo "Enter year:"
read year
if(($((year % 400)) -eq 0) || [(($((year % 4)) -eq 0 -a $((year % 100)) -ne 0)]
then
    echo "$year is a leap year"
else
    echo "$year is not a leap year"
fi
```

The bottom of the window shows the status bar with '-- INSERT --' on the left, '8,3' in the center, and 'All' on the right.

OUTPUT:

A terminal window titled 'user123@localhost:~' with search, menu, and close buttons. The terminal shows the following commands and output:

```
[user123@localhost ~]$ vi leap.sh
[user123@localhost ~]$ sh leap.sh
Enter year:
2006
2006 is not a leap year
[user123@localhost ~]$
```

RESULT:

Thus, the shell script to check leap year was executed successfully.

2. Shell Script for Arithmetic Operations

AIM:

To write and execute a shell script to perform arithmetic operations.

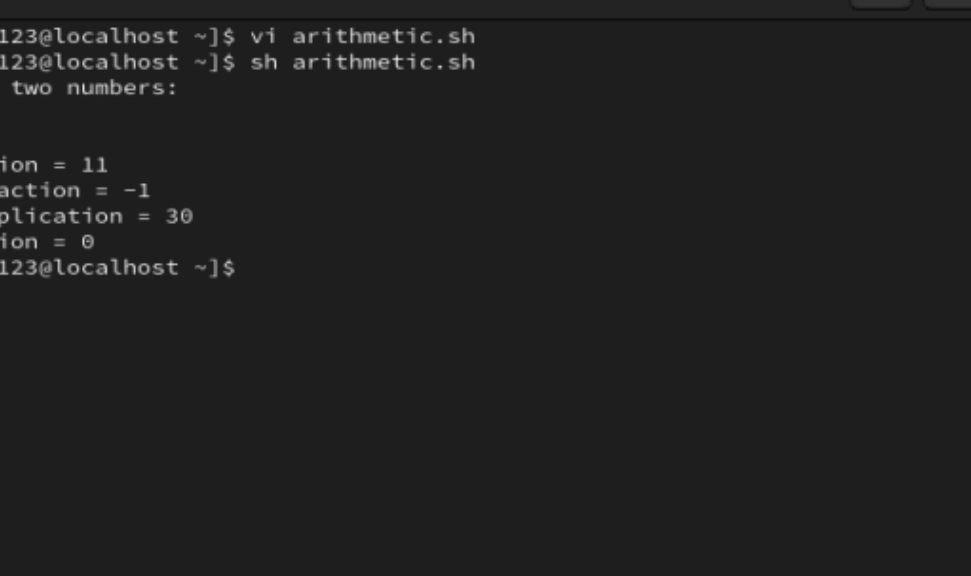
CODE:

```
user123@localhost:~ — /usr/bin/vim arithmetic.sh
```

```
.echo "Enter two numbers:"  
read a  
read b  
  
sum=$((a+b))  
sub=$((a-b))  
mul=$((a*b))  
div=$((a/b))  
  
echo "Addition = $sum"  
echo "Subtraction = $sub"  
echo "Multiplication = $mul"  
echo "Division = $div"
```

```
-- INSERT --
```

OUTPUT:



A terminal window titled "user123@localhost:~" with standard macOS window controls (search, menu, close). The terminal shows a user running a script named "arithmetic.sh" using "vi" and "sh". The script prompts for two numbers, 5 and 6, and then displays the results of four arithmetic operations: Addition = 11, Subtraction = -1, Multiplication = 30, and Division = 0. The prompt returns to the user's shell.

```
[user123@localhost ~]$ vi arithmetic.sh
[user123@localhost ~]$ sh arithmetic.sh
Enter two numbers:
5
6
Addition = 11
Subtraction = -1
Multiplication = 30
Division = 0
[user123@localhost ~]$
```

RESULT:

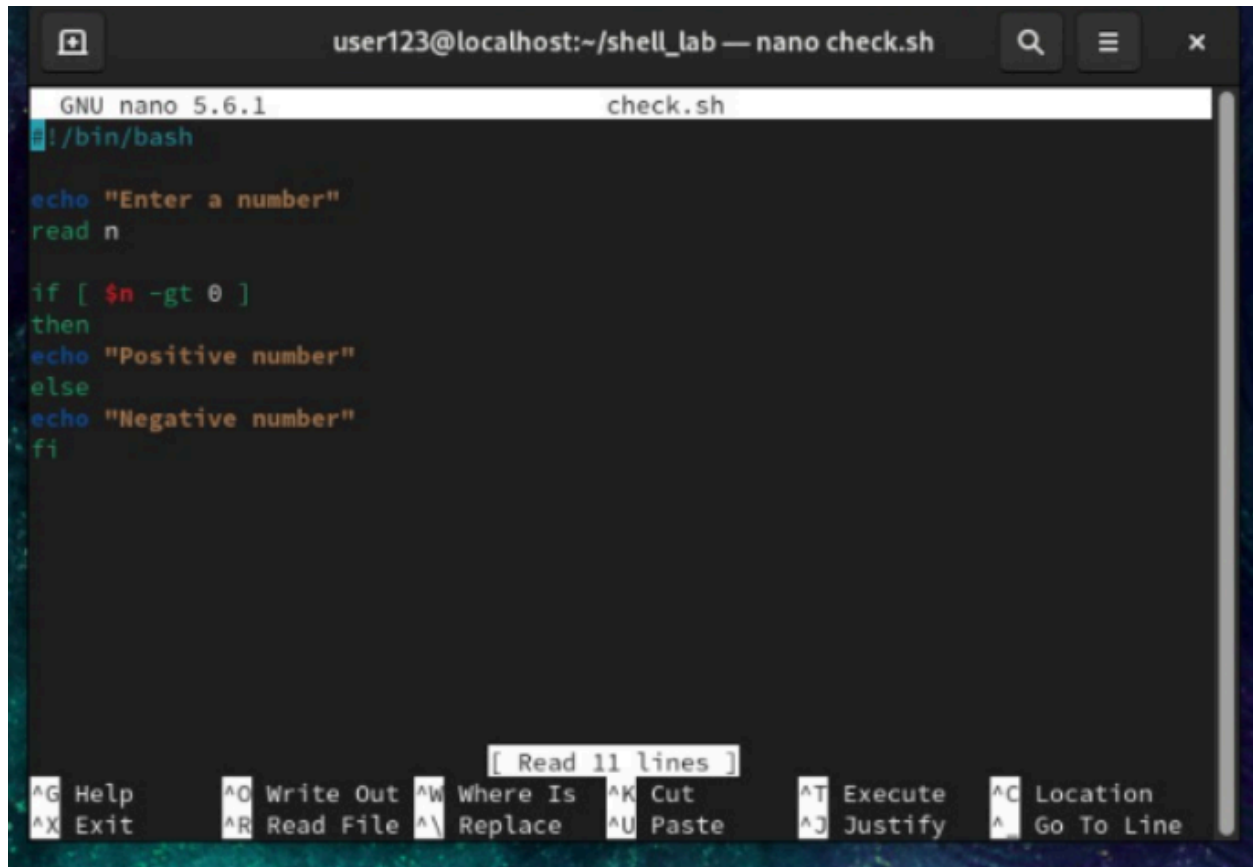
Thus, the shell script for arithmetic operations was executed successfully.

3. Shell Script for Positive or Negative numbers

AIM:

To write and execute a shell script to perform positive and negative numbers.

CODE:



```
GNU nano 5.6.1 check.sh
#!/bin/bash

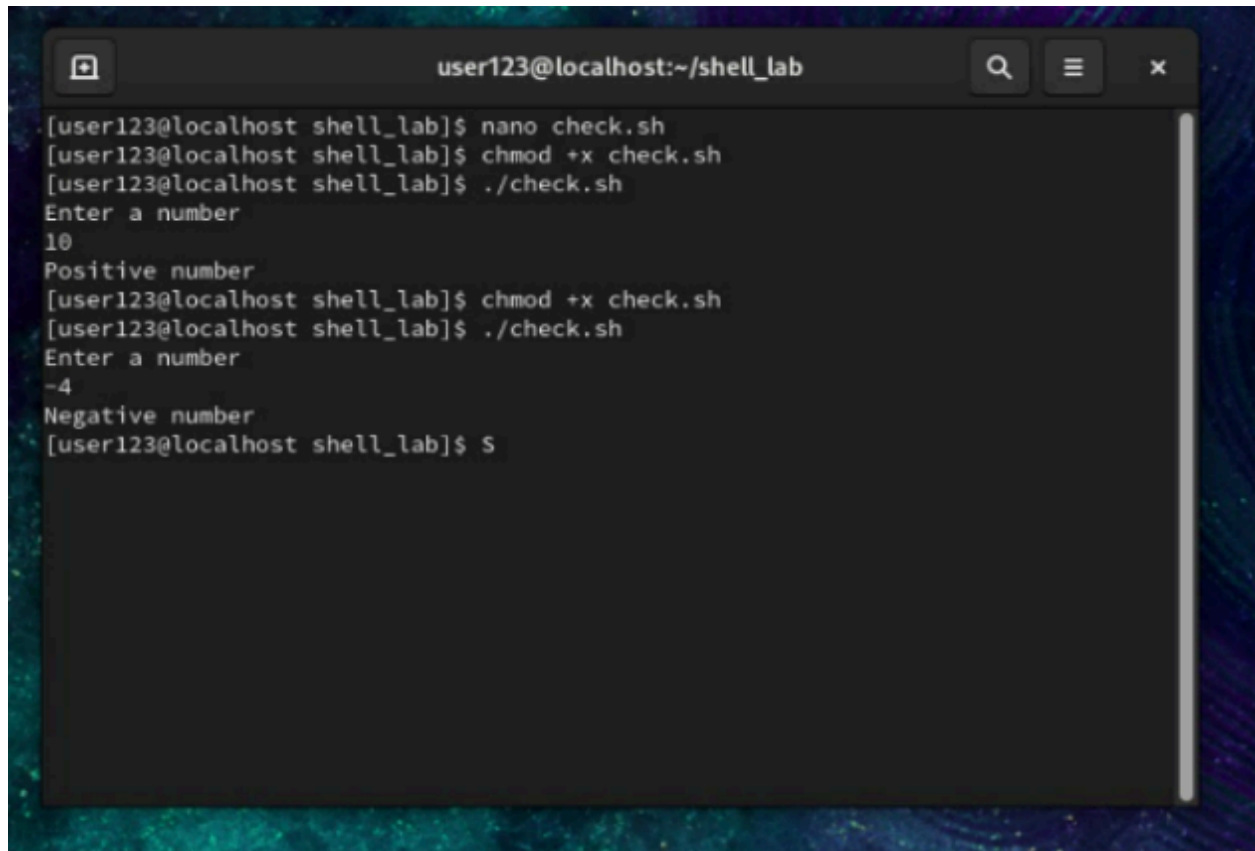
echo "Enter a number"
read n

if [ $n -gt 0 ]
then
echo "Positive number"
else
echo "Negative number"
fi
```

[Read 11 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^Y Replace	^U Paste	^J Justify	^_ Go To Line

OUTPUT:

A terminal window titled 'user123@localhost:~/shell_lab' with search, menu, and close buttons. It shows the execution of a shell script named 'check.sh'. The script prompts for a number. In the first run, '10' is entered, and the output is 'Positive number'. In the second run, '-4' is entered, and the output is 'Negative number'. The prompt returns to the shell.

```
[user123@localhost shell_lab]$ nano check.sh
[user123@localhost shell_lab]$ chmod +x check.sh
[user123@localhost shell_lab]$ ./check.sh
Enter a number
10
Positive number
[user123@localhost shell_lab]$ chmod +x check.sh
[user123@localhost shell_lab]$ ./check.sh
Enter a number
-4
Negative number
[user123@localhost shell_lab]$ s
```

RESULT:

Thus, the shell script for positive and negative number operations was executed successfully.